

18 Exponential and Poisson distributions

Subject content	Additional information
18.1 Determine when a Poisson model is appropriate (in real world situations including modelling assumptions).	Students will be expected to know the conditions necessary for a Poisson model to be appropriate and to explain or discuss whether these conditions are likely to be present in a given context.
18.2 Determine when an exponential distribution is appropriate (and its relationship to the Poisson distribution as a model of the times between randomly occurring Poisson events).	Students will be expected to know and use the relationship between corresponding Poisson and exponential distributions, in a given context, for Poisson events occurring randomly in time or in space.
18.3 Evaluate probabilities for Poisson and exponential distributions and know the corresponding mean and variance.	Students will not be expected to derive the formulae for the mean and variance.